Welcome to Colab!

Start coding or generate with AI.

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Default title text

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# @title Default title text
# Kitchen Vision AI (50 line program)

!pip install opencv-python tensorflow numpy matplotlib

import cv2
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt
from google.colab import files
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.applications.mobilenet_v2 import preprocess_input, decode_predicti

print("Loading AI Model...")
model = MobileNetV2(weights='imagenet')
print("Model Loaded")

print("Upload a kitchen image")
uploaded = files.upload()

def process_image(path):
    img = cv2.imread(path)
    img = cv2.resize(img,(224,224))
    arr = np.array(img)
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    arr = np.array(img)
    arr = np.expand_dims(arr,axis=0)
    arr = preprocess_input(arr)
    return img, arr

def detect_object(arr):
    pred = model.predict(arr)
    result = decode_predictions(pred, top=3)[0]
    print("\nDetected Items:")
    for r in result:
        print(r[1], "-", round(r[2]*100,2), "%")

for file in uploaded.keys():
    img, arr = process_image(file)
    detect_object(arr)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.title("Kitchen Vision Output")
    plt.axis("off")
    plt.show()

print("Program Finished")
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Loading AI Model...
Model Loaded
Upload a kitchen image
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Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

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Saving images.jpg to images (1).jpg
1/1 ━━━━━━━━━━━━━━━━━━━━ 2s 2s/step

Detected Items:
acorn_squash - 20.34 %
bell_pepper - 18.04 %
vase - 5.01 %
```

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Program Finished
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